

Electro-Optical Phase-Locked Loop Generating Linear Frequency Chirp for FMCW LiDAR

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Abstract: We demonstrated linear frequency-chirp modulation using a wide-bandwidth analog electro-optical phase-locked loop, leveraging linear analog multipliers in a single sideband frequency down-conversion scheme, without electrical ramp generation, predistortion circuitry, or complex digital logic. © 2020 The Author(s)

1. Introduction

A frequency modulated continuous wave (FMCW) light detection and ranging (LiDAR), based on linear frequency chirp, allows precise ranging with low optical power. The performance of FMCW LiDAR depends on the linearity of frequency chirp. Open-loop frequency modulation approaches, such as those relying on direct laser modulation, do not result in linear frequency chirps and require predistortion or other linearization methods. Feedback can be used to linearize the laser's frequency chirp modulation [1-3]. In this paper, we demonstrate the generation of linear frequency chirp by an electro-optical phase locked loop (EO-PLL). Compared to past EO-PLL-based FMCW generators, this scheme does not require external ramp generators, predistortion circuitry, or complex digital logic.

2. EO-PLL for Linear Frequency Chirp

Figure 1 shows the schematic of the EO-PLL. In general, the laser diode (LD) output field can be expressed as $E(t) = E_o \cos(\Omega_o t + \Phi(t))$. Ω_o is the laser's angular frequency and $\Phi(t)$ is its phase modulation. A delay-line-based Mach-Zehnder interferometer (MZI) and balanced photodiode (PD), acting as a frequency discriminator, follows the LD resulting in $V_{PD}(t) \propto \cos(\Omega_o \tau_{MZI} + \Phi(t) - \Phi(t - \tau_{MZI}))$. This signal is then compared with a reference electrical signal given by $V_{ref}(t) = V_o \cos(\omega_o t)$ using a current-mode single-sideband quadrature frequency down-converter, realized using linear analog multipliers (as opposed to switching mixers) resulting in $I_{beat}(t) \propto \cos(\Omega_o \tau_{MZI} + \Phi(t) - \Phi(t - \tau_{MZI}))$. This beat signal, after conversion to voltage through a differential transimpedance amplifier (TIA), goes through a differential integrator and feeds the laser through a current driver. In steady-state, the feedback requires $\Omega_o \tau_{MZI} + \Phi(t) - \Phi(t - \tau_{MZI})$ to be constant which implies that $\Phi(t) = \frac{\omega_o}{2 \times \tau_{MZI}} t^2$. In other words, in steady-state the feedback loop results in a linear frequency chirp generation whose rate is set by the frequency of the electrical signal source, ω_o , and the optical delay in the MZI, τ_{MZI} . The polarity and duration of the frequency chirp are set by a pair of switches placed between the differential TIA and the differential integrator.

We constructed the EO-PLL with discrete components mounted on a printed circuit board. The LD is a distributed feedback (DFB) laser with wavelength and output power of 1549 nm and 70 mW, respectively. The wavelength is tuned by a current injection with thermal effect. The LD output power does not change significantly as it is tuned due to the small injected current. In this prototype, τ_{MZI} is 15.0 ns and ω_o is 136 kHz, resulting in a chirp rate (ω_o/τ_{MZI}) of 9.06 GHz/ms. We switched the chirp direction at a period of 1.12 ms, thus total frequency change was 10.1 GHz. The beat spectrum at the balanced-PD output is shown in Fig. 2(a). When the LD is modulated only by a linear-current-ramp without EO-PLL, spectral peak was hardly seen because of the strong nonlinearity of the LD. When the EO-PLL is

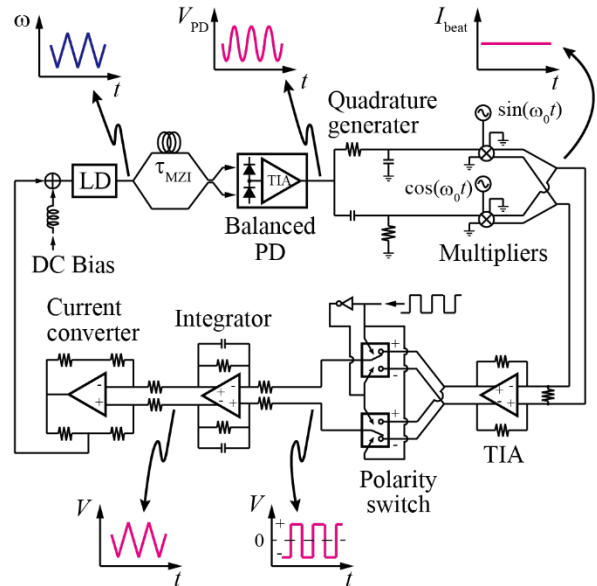


Fig. 1. Schematic of EO-PLL and signal waveforms at each point in stable condition.

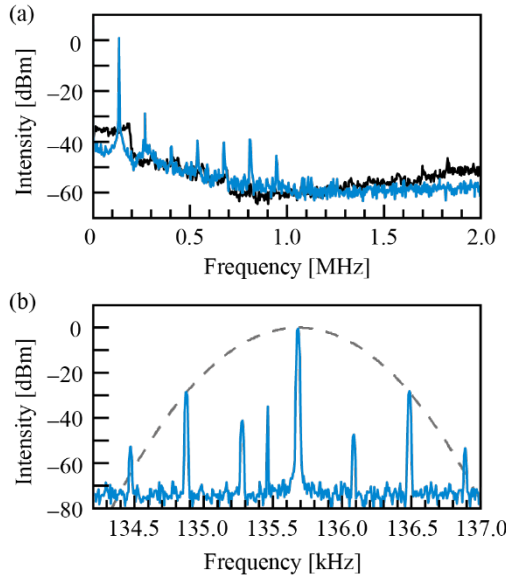


Fig. 2. Beat signal spectrum. (a) The spectra measured with the EO-PLL (blue) and only with a linear-current modulation of LD (black). (b) Magnification of the peak. Dotted line is Gaussian fitting of the peak.

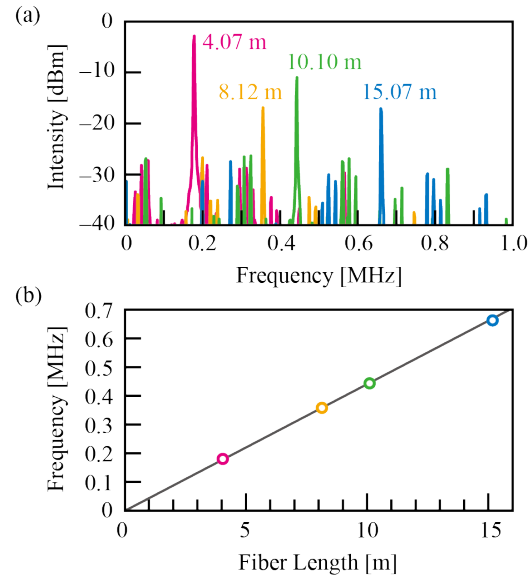


Fig. 3. Results of FMCW ranging. (a) Ranging spectra and (b) measured peak frequency for fiber length. Gray line shows the expected frequency from the chirp rate and the utilized fiber length.

locked, a sharp peak appeared at 136 kHz, which indicates that the linear chirp is successfully generated. The presence of harmonics is likely due to imperfections of the multiplier and the single sideband mixing, and can be eliminated by optimizing the circuitry. Figure 2(b) is a magnification of the peak measured with a resolution bandwidth of 10 Hz. The peak is split by the switching frequency of 890 Hz, and a full width at half maximum of a Gaussian fitting of the peak was 170 Hz.

To demonstrate the frequency chirp linearity, ranging experiments with precise lengths of fibers were conducted. Figures 3(a) and 3(b) show the ranging spectra measured with different fiber lengths, and the peak frequencies for each fiber length. The peak frequency linearly shifted along with the fiber length, confirming the chirp linearity.

3. Performance Comparison

The proposed EO-PLL is a fully analog architecture, and leads to linear frequency chirp generation without requiring external electrical ramp generators, predistortion circuitry, or complex digital logic. A wide loop bandwidth is achieved by using a linear analog multiplier (as opposed to a switching mixer) in a single-sideband frequency down-conversion scheme to eliminate the unwanted

harmonics without requiring loop filter with low cutoff frequency. In the demonstrated prototype, the DFB laser's slow response of the thermal wavelength tuning restricted the chirp rate. A distributed Bragg reflector (DBR) laser which can tune fast with carrier effects enables higher chirp rates [2].

4. References

- [1] N. Satyan et al., "Precise control of broadband frequency chirps using optoelectronic feedback," *Opt. Express*, vol. 17, no. 18, pp.15991–15999, 2009.
- [2] B. Behroozpour et al., "Electronic-photonic integrated circuit for 3D microimaging," *IEEE J. Solid-State Circuits*, vol. 52, no. 1, pp.161–172, 2017.
- [3] A. Binaie et al., "A 65nm CMOS continuous-time Electro-optic PLL (CT-EOPLL) with image and harmonic spur suppression for LIDAR," *IEEE RFIC Symposium*, pp.103–106, 2019.

Table 1. Comparison of performance of EO-PLL.

	Ref. [1]	Ref. [2]	Ref. [3]	This work
Architecture	Discrete	CMOS & Si photonics	CMOS & Si photonics	Discrete
Total Freq. Change	100 GHz	122 GHz	16 GHz	10 GHz
Chirp Period	1 ms	5.5 μ s	0.5 ms	1.1 ms
Chirp Rate	100 GHz/ms	22 THz/ms	32 GHz/ms	9 GHz/ms
Loop BW	200 kHz	700 kHz	20 kHz	100 kHz
Laser Type	DFB	DBR	DFB	DFB
External Ramp	Yes	No	Yes	No
Scheme	Analog	Digital	Analog	Analog